

~minute sheet ~

Inversion Lemmata

Inversion : Suppose $\Gamma \vdash e : T$

1. (PLUS case) If $e = \text{plus}(e_1; e_2)$, then it must be that

- $T = \text{Num}$
- $\Gamma \vdash e_1 : \text{Num}$
- $\Gamma \vdash e_2 : \text{Num}$

$$\text{CAT} \frac{\Gamma \vdash e_1 : \text{Str} \quad \Gamma \vdash e_2 : \text{Str}}{\Gamma \vdash \text{cat}(e_1; e_2) : \text{Str}}$$

2. If $e = \text{cat}(e_1; e_2)$ then it must be that

- $T = \text{Str}$
- $\Gamma \vdash e_1 : \text{Str}$
- $\Gamma \vdash e_2 : \text{Str}$

Proven by:

- Induction
- Inspection

Weakening

Weakening: if $\Gamma \vdash e : \tau$ and x is fresh
then $\Gamma, x : \sigma \vdash e : \tau$

Intuition:

$x : \text{Num} \vdash \text{plus}(\text{num}[3]; \text{num}[2]) : \text{Num}$

$$\frac{\frac{3 \in \mathbb{N}}{x : \text{Num} \vdash \text{num}[3] : \text{Num}}^{\text{Num}} \quad \frac{2 \in \mathbb{N}}{x : \text{Num} \vdash \text{num}[2] : \text{Num}}^{\text{Num}}}{x : \text{Num} \vdash \text{plus}(\text{num}[3]; \text{num}[2]) : \text{Num}}^{\text{PLUS}}$$



$$\frac{\frac{3 \in \mathbb{N}}{x : \text{Num} \vdash \text{num}[3] : \text{Num}}^{\text{Num}} \quad \frac{2 \in \mathbb{N}}{x : \text{Num} \vdash \text{num}[2] : \text{Num}}^{\text{Num}}}{x : \text{Num} \vdash \text{plus}(\text{num}[3]; \text{num}[2]) : \text{Num}}^{\text{PLUS}}$$

Note: In the original image, there are blue annotations 'y:G' above the 'x' in the premises and below the 'x' in the conclusion of the second derivation.

Substitution

Meta-theoretic ops = operations that we will use in the specification of our language that are not constructs of our language.

$$z[e/x] = \begin{cases} e & \text{if } z \equiv x \\ z & \text{if } z \neq x \end{cases}$$

$$(\text{num}[n])[e/x] = \text{num}[n]$$

$$(\text{str}[s])[e/x] = \text{str}[s]$$

$$(\text{plus}(e_1, e_2))[e/x] = \text{plus}(e_1[e/x], e_2[e/x])$$

Times, cat and len are analogous

Example

$$(\text{plus}(\text{num}[1], x)) [\text{num}[2]/x]$$

$$\rightarrow \text{plus}(\text{num}[1] [\text{num}[2]/x], x [\text{num}[2]/x])$$

$$\rightarrow \text{plus}(\text{num}[1], x [\text{num}[2]/x])$$

$$\rightarrow \text{plus}(\text{num}[1], \text{num}[2])$$

$$[\text{num}[2]/y]$$

$$\rightarrow \text{plus}(\text{num}[1], x)$$

Let

$$(\text{let}(e_1; y. e_2)) [e/x] = \text{let}(e_1 [e/x]; y. e_2 [e/x])$$

$$\text{let}(y; y. \text{len}(y)) [\text{str}[n]/y]$$

↑ ↑
not the same y
→ referential clash!

$$\text{let}(x; \underline{y. x + y}) [y/x]$$

↑ ↑
Free!
if y is shadowed in here for x, it will lose its freedom!

→ variable capture

Barendregt Convention

= we assume that everything has been alpha renamed already so that this doesn't happen.

Substitution: If $\Gamma \vdash e : T$ and (lemma) $\Gamma, x : T \vdash u : \sigma$

then $\Gamma \vdash u[e/x] : \sigma$

Proof by induction on $\Gamma, x : T \vdash u : \sigma$.

[Case: LET]

GOAL = if $\Gamma \vdash e : T$

and $\Gamma, x : T \vdash \text{let}(e_1; y. e_2) : \sigma$

then

$\Gamma \vdash (\text{let}(e_1; y. e_2)) [e/x] : \sigma$

$P1 = \Gamma \vdash e : T$

$P2 = \Gamma, x : T \vdash \text{let}(e_1; y. e_2) : \sigma$

GOAL' = $\Gamma \vdash (\text{let}(e_1; y. e_2)) [e/x] : \sigma$

By inversion, $P2$ must have had the following derivation with sub derivations we named for later use

$$\frac{\begin{array}{c} \vdash \text{SHAPE1} \\ \Gamma, x : T \vdash e_1 : \sigma_1 \end{array} \quad \begin{array}{c} \vdash \text{SHAPE2} \\ \Gamma, x : T, y : \sigma_1 \vdash e_2 : \sigma \end{array}}{\Gamma, x : T \vdash \text{let}(e_1; y. e_2) : \sigma} \text{LET}$$

IH1 = if $\Gamma \vdash e : T$ and $\Gamma, x : T \vdash e_1 : \sigma_1$ then $\Gamma \vdash e_1 [e/x] : \sigma_1$

$\neq P1$ Weaken $P1$

IH2 = if $\Gamma, y : \sigma_1 \vdash e : T$ and

$\Gamma, y : \sigma_1, x : T \vdash e_2 : \sigma$

then $\Gamma, y : \sigma_1 \vdash e_2 [e/x] : \sigma$

complete yourself + compare to the notes / BP file.